



Sara Varela

last seen yesterday at 11:07



¡Hola Lucas! We are organising an "Ecology meets Evolution" session at the joint AEET/SIBECOL congress (Pontevedra, 06/2025). Would you like to present something?

09:37

¡Buenas Sara! Well, I have some stuff going on but it's rather exploratory, and still preliminary. Is that okay?

09:38

¡Vaya, genial! What would you like to talk about?

09:38

Well, I think my title would be something like

09:39

Continent-scale insights into the last 23 million years of South American mammal evolution: importance of (a)biotic factors

Lucas Buffan^{i,*}, Myriam Boivinⁱⁱ, Fabien L. Condamineⁱ, Narla S. Stutzⁱ, François Pujosⁱⁱⁱ, Pierre-Olivier Antoineⁱ, Laurent Marivauxⁱ

09:40

If I might add a piece of context

09:40

- South America is a hotspot for mammal biodiversity, but South American mammal (SAM) communities have undergone important changes in the geological past¹
- In macroevolution, two nonmutually-exclusive hypotheses are generally invoked to describe how external factors shape biodiversity patterns²



"Red Queen" hypothesis: biodiversity patterns are mostly the product of **biotic interactions**



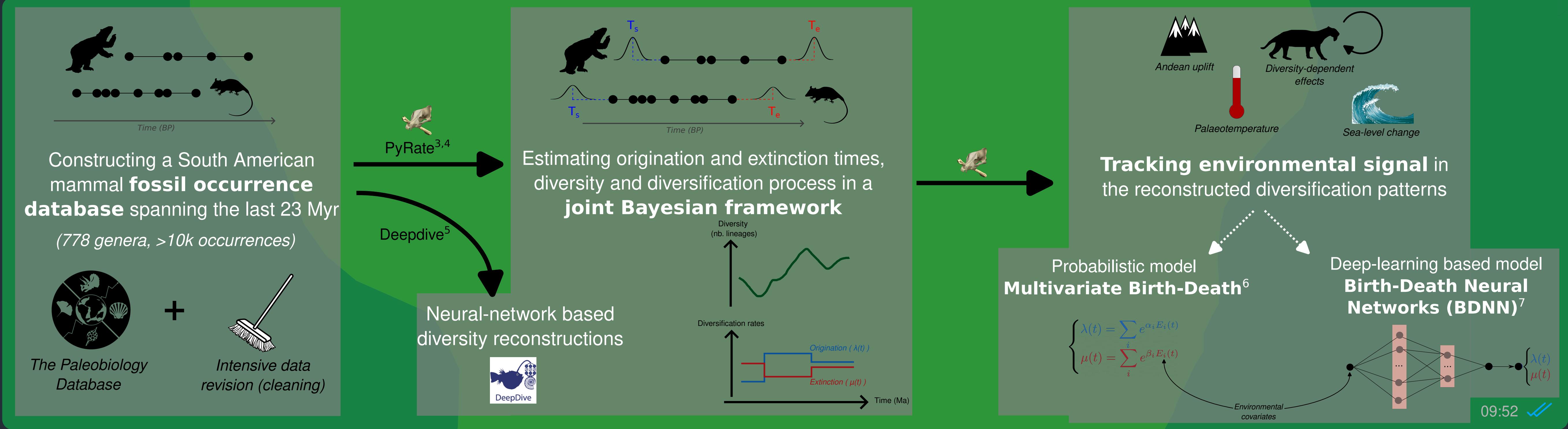
"Court Jester" hypothesis: biodiversity patterns are mostly driven by **abiotic factors**

- The **aim** of this study is to **reconstruct SAM macroevolutionary dynamics** throughout the last 23 My and to investigate their **likely environmental drivers**

09:44

Lucas, this is a broad topic. How would you approach it in practise?

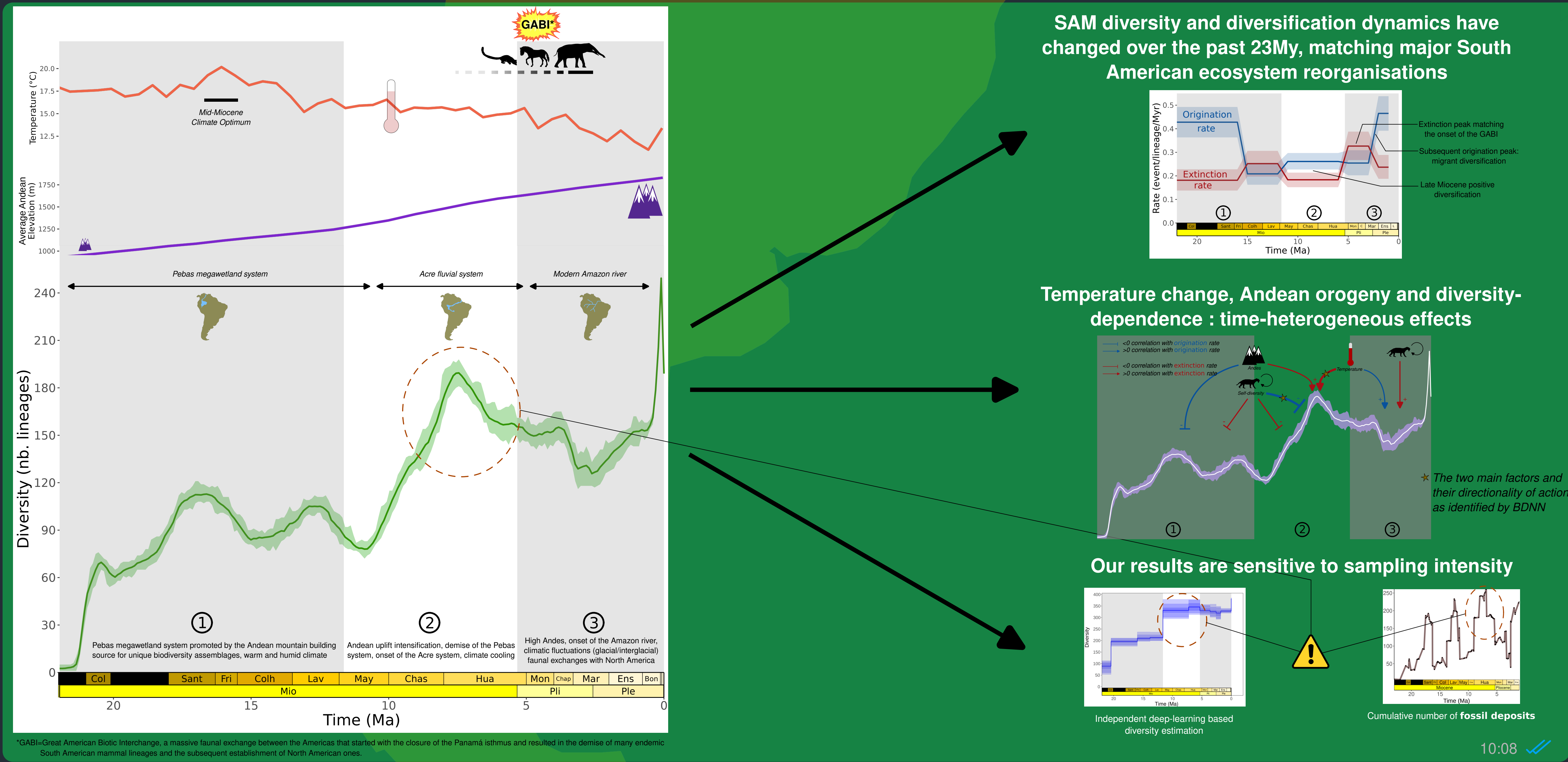
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09:52

Ok, sounds exciting! Any intersting finding(s) so far?

09:59



10:08

In a near future, the plan is to extract clade-specific diversity trajectories by incorporating inputs from molecular phylogenies to investigate the role of biotic interactions at a finer scale. Also, it would be interesting to decipher patterns by latitude, differentiating changes among tropical and extratropical faunas, and by altitude, differentiating the pace of evolution of low and high latitude taxa.

10:11

Bueno, I think you'll have some nice content to share, looking forward for hearing more about it! ¡Un abrazo!

10:21

Type a message



REFERENCES

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4. D. Silvestro, N. Salamin, A. Antonelli, X. Meyer, Improved estimation of macroevolutionary rates from fossil data using a Bayesian framework. *Paleobiology* 45, 546–570 (2019)
5. R. B. Cooper, J. T. Flannery-Sutherland, D. Silvestro, DeepDive: estimating global biodiversity patterns through time using deep learning. *Nat. Comm.* 15, 4199 (2024)
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AFFILIATIONS

- i. Institut des Sciences de l'Évolution de Montpellier (ISEM), Université de Montpellier, CNRS, IRD, Montpellier cedex 5 34095, France
- ii. Grupo de Paleontología de Vertebrados, Instituto de Ecorregiones Andinas (INECOA), Universidad Nacional de Jujuy (UNJU), CONICET–San Salvador de Jujuy, Argentina
- iii. Instituto Argentino de Nivología, Glaciología y Ciencias Ambientales (IANIGLA), Centro Científico Tecnológico, CONICET–Mendoza, Parque General San Martín, Mendoza 5500, Argentina

*✉ lucas.buffan@hotmail.fr @lucasbuffan.bsky.social

